

HATCHERY BUILDING INSULATION & CEILINGS REPLACEMENT PROJECT

Jordan River National Fish Hatchery
Antrim County, Michigan

SPECIFICATIONS



United States Department of the Interior
Fish and Wildlife Service
Infrastructure Management Division
Region 3, Bloomington, MN 55437-1458

Some blank pages are intentionally included.
Specifications are setup for two sided printing.

PART III, SECTION J

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PART 1 - GENERAL

- 1.1 Project Title: Hatchery Building Insulation and ceilings replacement project at the Jordan River National Fish Hatchery.
- 1.2 Work under these Specifications: The Fish & Wildlife Service is requesting bids to provide all labor, materials, equipment, project coordination, mobilization, demobilization, and submittals required to remove and replace insulation and ceilings in the existing hatchery building. The existing attic insulation is supported by plastic sheeting and is partially falling down onto the existing drop ceiling in some areas. This project would remove the existing insulation and install new drywall ceilings to support the new insulation. Additional work to replace the acoustical drop ceiling and lighting is also included as per the drawings. The project is broken down into a base bid for the office area of the building and bid additive #1 for the employee break room, hallway, dry storage, and bathroom area of the building on the very south end. The bid additive will be included if funds are available.

The contractor shall furnish all materials, labor, and equipment for the work further described below.

Work includes but is not limited to:

1. Removal, disposal, and clean-up of existing attic insulation where specified on the drawings. The existing plastic sheeting that holds up the attic insulation has fallen down in areas. Insulation to be removed is also on the existing drop ceiling in some offices.
2. Removal and disposal of existing plastic sheeting, acoustical drop ceiling, hangers, lighting, and electrical in preparation for new ceilings and insulation.
3. Installing new 5/8" gypsum board ceiling and sealing all penetrations into the attic for areas of the building that previously had plastic holding up the attic insulation.
4. Installing new R-49 insulation across the entire attic. Including areas of existing

- hard ceiling and new gypsum board ceiling. Also includes installing new attic insulation vent baffles to allow for new insulation depth.
5. Installing new acoustical ceiling, grid, and LED lighting troffers in areas specified. Also, includes modifying existing electrical wiring to install new lighting.
 6. Other miscellaneous items as shown on the drawings and as required by the specifications.

This work shall not be measured for payment and shall be paid for on a lump sum basis, complete.

1.3 Location and Inspection of Site

- A. The work site is located at the Jordan River National Fish Hatchery near Elmira Michigan as shown on the included drawings. The mailing address is shown on the Solicitation, Offer, and Award (Standard Forms 1442 or, 18 block 9) as attached to this Invitation.
- B. All prospective bidders/quoters, contractors, subcontractors, and related persons are encouraged to visit the site(s) of the proposed work and to assess the location(s) and extent of the work prior to submitting bids/quotes. Persons may inspect the work site by contacting the Hatchery Manager. All questions regarding the scope of work shall be presented in writing to the Contract Officer.

1.4 Specifications and Drawings for Construction

- A. The successful bidder will be provided, upon request, pdf or AutoCAD versions of the contract drawings as necessary for execution of this contract. These drawings will be provided at no additional cost to the successful bidder/quoter.
- B. Exact location of all affected utilities will be the responsibility of the contractor. Drawings do not completely show existing utilities. Site verification is required in addition to contacting any local authorities and utility

companies to determine exact locations of utilities.

- C. In case of difference between drawings and specifications, the specifications shall govern.

1.5 Work Not Included In This Contract

- A. All items indicated "**NIC**" on drawings or in specifications are items not included in this contract.
- B. Work in place is generally shown as "Existing" on the Contract Drawings.
- C. Work to be performed under other contracts and not included as part of this contract is generally shown as "By Others" on the Contract Drawings.

1.6 Items Provided By The Government

- A. Work to be accomplished by the Government or manufactured items to be provided by the Government are generally shown as "By Government" on the Contract Drawings.
 - (a) Unless specifically provided herein, no materials or equipment shall be furnished by the Government under this Contract.

1.7 Temporary Facilities

- A. Maintain and restore all Government roads, service areas and utilities used to pre-construction condition.
- B. Temporary Toilet facilities will be required for workers throughout the project.
- C. Large amounts (>100gallons) of water needed for construction shall be provided by the contractor.
- D. The contractor may use the existing electrical /power supply for small construction activities during this project. Large amounts (>5KWH per

day) of electrical power will be the responsibility of the contractor.

- E. The hatchery will not provide phone/fax services and will not accept deliveries or unload deliveries.
- F. The hatchery will not provide tools, parts, or other equipment to the contractor for use during construction.

1.8 Site Access, Staking and Right of Way

- A. The construction access limits on Government property will be established by the Government and the Contractor shall be responsible for complying with these limits and governing all work accordingly. All temporary access roads shall be removed following work completion unless directed otherwise by the Contracting Officer.
- B. Construction staking will not be provided by the Government, the contractor shall include all costs to stakeout, mark, survey, or otherwise layout the work to be accomplished as part of this contract.
- C. Maintain traffic and public safety in a safe manner with signs, barricades, lights, and watchmen as required. Follow Manual of Uniform Traffic Control Devices (MUTCD). Load limits and other restrictions shall be obeyed. All bidders are reminded that they are responsible for contacting the local entity such as township, county, or state having jurisdiction in the project vicinity to insure bidders/quoters are aware of any requirements of the local entity. These requirements are between the entity and the bidder. If the Contractor proposes to cross private property, permission must be obtained from the owner to use private property or roads in the conduct of the Work.
- D. Work scheduled on Saturdays, Sundays, Federal Holidays or other than normal working hours shall be approved in advance by the COR, Hatchery Manager, and Contracting Officer.

1.9 Safety - In addition to FAR 52.236-13 Accident Prevention

- A. The contractor shall designate a safety officer to ensure a safe work site. The safety officer shall perform daily work site inspections to assure compliance with all safety regulations and standards.
- B. The Contractor's Safety Plan shall be available on-site. The Contractor shall provide Safety Data Sheets (SDS) for all hazardous materials brought on-site. SDS shall be made available in a conspicuous location.
- C. The Contractor shall provide a Pre-construction Safety Checklist during the preconstruction conference.
- D. All equipment and work activities shall comply with OSHA Regulations and all accidents incidental to work performed under this contract shall be reported in accordance with OSHA reporting requirements. Additionally a copy of the accident report shall be provided to the Contracting Officer.
- E. The contractor shall provide and install safety barricades, signs, or other warning devices to delineate & warn the public of the construction area.
- F. Provide at least one non-freezing-type fire extinguisher in each workshop and shed used for storage of materials on the premises. Place in location readily accessible to workers.
- G. Provide at least one non-freezing-type fire extinguisher of the appropriate class for each group of workers performing flammable procedures.
- H. Provide and maintain a basic first aid kit for use of workers.
 - (a) Provide first aid supply commensurate with size of project with items necessary for

first aid treatment of all injuries. The first aid supply shall include items identified on SDS (see paragraph B of this section) in applicable locations.

- (b) Advise all workers of the location of first aid supplies.
 - (c) Post telephone numbers of nearest hospital or ambulance service and fire station in conspicuous location. Advise all workers of location of telephone numbers.
- I. Provide necessary Personnel Protective Equipment (PPE) as identified by SDS and OSHA regulations. Provide documentation, workers have been trained in use of PPE prior to individual use.

1.10 Environmental Requirements

- A. The contractor shall not use any products containing asbestos, cadmium, chromium, or lead. The contractor shall be responsible for removing and replacing materials found to contain these items at their own expense.

(END OF SECTION 01100)

PART 1 - GENERAL

1.1 Jobsite Administration - Superintendence, Subcontract administration and Quality Control, is the responsibility of the Contractor.

- A. Final decisions relative to items in conflict shall be made by the Contracting Officer.
- B. The Government shall have the authority to temporarily suspend the Work for such time as necessary because of the failure on the part of the Contractor to perform any of the provisions of the Contract. No extension of contract performance time will be permitted until the Government determines that the deficiencies in performance, which caused the temporary suspension, have been corrected.
- C. The Contractor shall have responsible superintendence on site during all project operations. Contractor shall notify Government of any changes of superintendents.
- D. The Contractor shall notify the COR at least ten (10) days prior to the anticipated date of completion of all Work specified herein. Upon completion of the Work, the COR shall proceed with the final inspection which shall be completed as promptly as practicable. The time required for such inspection and for performing any corrective Work shall be included in the contract performance time.
- F. The contractor shall notify the COR and Inspector five (5) days prior to any concrete pours, paving and back filling so that the Inspector can be on-site to perform inspections.

1.2 Modification/Change Order procedures

- A. In the event of the need for a Change Order, the Contractor will receive a letter requesting a price quote for proposed changes to the Contract.

- B. For evaluation purposes, the Contractor's quote shall be broken down to show the costs of labor and materials for each proposed category of Work included with the change, along with the total cost for Contractor's overhead, profit, and bond for the proposed changes.
- C. All Contract time extensions required as a result of a proposed change must be justified and supported in detail at the time of the proposal.

1.3 Preconstruction Conference

- A. Soon after award of the Contract and prior to commencing construction, the Contractor and any subcontractors shall attend a preconstruction conference with representatives of the Government. Timely notification of the meeting will be given to the Contractor. Time, date, and place of the conference will be determined by the Contracting Officer. Requirements pertaining to Contract administration, construction operations and the project schedule will be discussed. The project schedule shall be in accordance with Section 01330. The Preconstruction Safety Checklist shall be submitted for review during this conference. Please pay special attention to Question #15 of the "Preconstruction Safety Checklist".

1.4 Progress Meetings

- A. Periodic general project meetings will be conducted with a representative of the Contractor at the project site or by telephone by the Government's representative for the purpose of coordinating, expediting, and assessing the work progress.
- B. The Contractor shall provide reports of progress on the project, discuss the work schedule for the coming period and present all conflicts, discrepancies or other difficulties for resolution. A record of

DIVISION 1
GENERAL REQUIREMENTS

SECTION 01310
PROJECT COORDINATION

discussions and decisions will be prepared by the Government and distributed to those attending and others affected.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (not Applicable)

(END OF SECTION 01310)

PART 1 - GENERAL

1.1 General:

- A. Refer to submittal register or technical specifications for all submittals required.
- B. When specifically requested by the Contracting Officer (CO), Contractor's Officer's Representative (COR) or the Government's Construction Inspector, the Contractor shall provide copies of manufacturer's written installation instructions and/or recommendations for review and approval prior to accomplishing equipment installation.

1.2 Submittal Register:

- A. A submittal register, summarizing (not a complete listing) major items requiring submittals under this solicitation, is provided as an attachment to this document. The Contractor shall provide all required submittals to the COR within 30 days of award except for items noted otherwise on the submittal register.
- B. Furnish a minimum of four copies of all required submittals. Three copies will be retained by the Government. The remaining copies will be returned to the Contractor.
- C. The approval of submittals shall not relieve the Contractor to comply with the terms and conditions of the Contract Drawings and Specifications. The Contractor shall be responsible to determine correct dimension, design, connection, detail and satisfactory construction of all Work.

1.3 Project Schedule: Contractor shall provide as required by Federal Acquisition Regulation (FAR) clauses. *

1.4 Price Breakdown - Lump Sum Items: Contractor shall provide as required by FAR clauses. *

1.5 Subcontractor's List

A. Submit a list of subcontractors who will be working on the project.

1.6 Submittal of Progress Payment Certificate: Contractor shall provide as required by FAR clauses. *

1.7 Guarantees, Warranties And Certificates: Refer to Section 01770

1.8 Operating And Maintenance Instruction: Refer to Section 01770

1.9 Delivery Tickets

A. The Contractor shall submit to the Government's on site representative, one legible copy of each delivery ticket for all material delivered to the construction site.

B. The delivery ticket shall show brand name, catalog number, and number of items received.

1.10 Waste Manifests/Load Tickets

A. The Contractor shall submit copies of all waste manifests and/or load tickets.

B. Manifests and or load tickets shall state type of waste (i.e., construction debris, hazardous asbestos, etc.), location of disposal, landfill designation, EPA registration of landfill, and EPA registration for transportation of any hazardous wastes.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (not Applicable)

DIVISION 1
GENERAL REQUIREMENTS

SECTION 01330
SUBMITTALS

- * The applicable FAR clauses will be provided in full text to the resultant contractor at the pre-construction conference.

(END OF SECTION 01330)

PART 1 - GENERAL

1.1 Regulatory Requirements

- A. All Work shall be accomplished in strict accordance with the regulations, codes, and laws in effect at the time the Work is performed for the location where it is to be performed. All completed construction shall comply with OSHA, NFPA, NEC and ABA regulations.
- B. All work performed on this project shall comply with Title 40, Protection of the Environment, Code of Federal Regulations.
- C. All electrical installations, components and work shall comply with the most current edition of NFPA 70 National Electric Code.

1.2 Means and Methods

- A. Means and methods used in the performance of this Contract must be in accordance with all applicable Federal Acquisition Regulations in addition to OSHA regulations, 29 Code of Federal Regulations, acceptable industry practices as recognized by societies, associations, building codes, manufacturers' specifications and any applicable state safety and health requirements.

1.3 Permits

- A. All permits required to support the construction activities such as road permits for oversize, overweight vehicles and equipment, bridge permits, and utility hookup permits, shall be obtained and paid for by the Contractor.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (not Applicable)

(END OF SECTION 01420)

PART 1 - GENERAL

1.1 Manufacturer's Specifications

- A. The installation of manufactured products and equipment, together with all associated parts, shall be in conformance with the manufacturer's recommendations, and code requirements. All installations shall be complete and ready for immediate use, and every part required for proper operation of the manufacturer's equipment shall be provided. Following the completion of the Work under this Contract, all installed equipment shall be tested for proper operation and compliance with the Contract.

1.2 Final Inspection

- A. The Contractor shall request a final inspection in writing at least ten (10) days prior to the anticipated date of completion.
- B. Work will not be considered ready for final inspection until all the Work has been completed and the Contractor has certified that all items are properly operating and in strict compliance with all Contract terms and conditions.
- C. The Contractor or the project supervisor shall be at the job site during the final inspection.

1.3 Tests

- A. Complete all tests required to prove actual operating performance of equipment and systems incorporated into the project.
- B. Submit reports of all tests to the Contracting Officer prior to final payment.
- C. Provide any regulatory acceptance documents in cases of cleanup.

1.4 Maintenance and Operating Instructions

- A. Submit three (3) final copies of all operating, maintenance instructions and parts lists for all equipment and systems that require adjustment, servicing, and attention for proper operation. Include all information necessary for proper operation of equipment. Use simple, non-technical language where possible. Provide all necessary diagrams and schematics required by repairmen. Any hazards likely to occur in operation of equipment shall be pointed out to avoid improper operation of any equipment that might result in damage or danger to equipment, building or personnel.
- B. Final copies of all maintenance and operating information shall be submitted in bound form as a "Manual of Operating Instructions" with items grouped by sections of Work. Append each manual with name, address, telephone number of Contractor, subcontractor and companies or individuals providing service for equipment used.
- C. Provide competent personnel to meet with the Government representative to fully explain and familiarize the Government personnel with all equipment and its operation and maintenance.

1.5 Record Drawings:

The Contractor shall provide and maintain in proper order and in good, clean condition, at the project office or at the project site, one (1) complete set of full size prints of all Drawings. On this set, with red pencil or ink, the Contractor will neatly inscribe all changes in Work. At time of final acceptance, and prior to final payment, these corrected prints will be presented to the COR. The Contractor will note all data and changes on these record drawings in sufficient detail and clarity to provide information necessary for preparation of "as built" drawings by the Government.

1.6 Guarantees, Warranties and Certificates

- A. The Contractor will submit all guarantees, warranties and certificates to the Government prior to final payment.
- B. The Government requires a 100% warranty for a period of one full year after final acceptance. Items of Work found defective during the warranty period will be replaced at no cost to the Government and the warranty for that particular part of the Work shall automatically extend for one full year beyond the completion date of repairs.
- C. If a manufacturer or supplier offers a guarantee/warranty or is required per the detailed specification to provide a guarantee/warranty in excess of the one year requirement outlined herein, then the longer period will apply for those particular items.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (not Applicable)

(END OF SECTION 01770)

PART 1 - GENERAL

1.1. General

- A. The work of this section consists of the demolition and removal of insulation, ceilings, and other miscellaneous items as shown on the contract drawings.

PART 2 - PRODUCTS

PART 3 - EXECUTION

3.1 Disposal

1. The contractor shall have a Waste Reduction Plan (FAR Clause 52.223.10) for demolition and construction material to divert as much material as feasibly possible to recycling or reuse.
2. Non-regulated demolition material shall be removed from the hatchery site and disposed of in accordance with state and local regulations.
2. All hazardous demolition material such as asbestos products, treated wood, or asphalt roofing shall be removed from the hatchery site and disposed of in an EPA licensed landfill or as otherwise required by state and local regulations.
3. The contractor is encouraged to take any recyclable materials removed from this project to a recycling center as appropriate.

3.2 Execution

1. The existing attic insulation has been tested and does not include asbestos materials. The contractor shall still remove the insulation in a manner that does not allow the insulation to become air-borne and get into the HVAC systems, offices, garages, or spread to other areas of the building. Temporary plastic sheeting, temporary curtain walls, and other isolation methods shall be

performed as required to not spread the dust/materials.

2. The contractor shall be required to replace and dispose of all HVAC filters at the completion of all work.
3. The contractor shall save existing smoke detectors, emergency lights, and other drop-ceiling mounted items for re-installation on new ceilings.
4. The Contractor shall shut off all electrical circuits while doing demolition. Install temporary boxes/covers on the ends of existing electrical wiring for lighting and other equipment so that the electric for the remaining part of the building can still be used during construction.
5. The Contractor shall erect temporary supports as needed in order to do any demolition required. This may included but is not limited to the existing ductwork, piping, and electrical.
6. The Contractor shall protect and clean all existing surfaces, carpets, and other office equipment that remains during construction.
7. The Contractor shall take precautions to insure that the remaining structures are not damaged. Any damage to the structure shall be repaired at the Contractor's expense.
8. Areas disturbed by the demolition which are not incorporated into the new work shall be restored to their original condition or better, even if the original condition was poor.

(END OF SECTION 02071)

SECTION 072100 - THERMAL INSULATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Blown-in cellulose insulation
- 2. Glass-fiber blanket insulation.
- 3. Vapor retarders.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

1.4 INFORMATIONAL SUBMITTALS

- A. Product Test Reports: Based on evaluation of comprehensive tests performed by a qualified testing agency, for each product.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Protect insulation materials from physical damage and from deterioration due to moisture, soiling, and other sources. Store inside and in a dry location. Comply with manufacturer's written instructions for handling, storing, and protecting during installation.

PART 2 - PRODUCTS

2.1 BLOWN IN CELLULOSE INSULATION

- A. Blown in insulation meeting the testing requirements of ASTM C-739, E-84 and E-119 and UL-723. Minimum 80% recycled wood-based cellulosic fibers with a minimum of 75 percent total recovered material content per EPA/CPG guidelines.
1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. US GreenFiber, LLC.
 - b. Nu-Wool Company, Inc.
 - c. National Fiber.
 2. Low Dust, Formaldehyde-free.
 3. R-Value of 3.6 per inch, minimum
 4. All material shall be asbestos free.

2.2 GLASS-FIBER BLANKET INSULATION

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. CertainTeed Corporation.
 2. Guardian Building Products, Inc.
 3. Johns Manville.
 4. Knauf Insulation.
 5. Owens Corning.
- B. Unfaced, Glass-Fiber Blanket Insulation: ASTM C 665, Type I; with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively, per ASTM E 84; passing ASTM E 136 for combustion characteristics.
- C. Sustainability Requirements: Provide glass-fiber blanket insulation as follows:

1. Free of Formaldehyde: Insulation manufactured with 100 percent acrylic binders and no formaldehyde.
2. Low Emitting: Insulation tested according to ASTM D 5116 and shown to emit less than 0.05-ppm formaldehyde.
3. All material shall be asbestos free.

2.3 ATTIC BAFFLES

- A. Provide Durovent attic baffles by ADO Products or approved equal.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean substrates of substances that are harmful to insulation or that interfere with insulation attachment.

3.2 INSTALLATION, GENERAL

- A. Comply with insulation manufacturer's written instructions applicable to products and applications indicated.
- B. Install insulation that is undamaged, dry, and unsoiled and that has not been left exposed to ice, rain, or snow at any time.
- C. Extend insulation to envelop entire area to be insulated. Cut and fit tightly around obstructions and fill voids with insulation. Remove projections that interfere with placement.
- D. Provide sizes to fit applications indicated and selected from manufacturer's standard thicknesses, widths, and lengths. Apply single layer of insulation units to produce thickness indicated unless multiple layers are otherwise shown or required to make up total thickness.

3.3 INSTALLATION OF ABOVE CEILING INSULATION

- A. On attic surfaces, blow insulation according to manufacturer's written instructions.

1. If not otherwise indicated, extend insulation the full width of the building.
 - B. On sloped ceilings, provide blocking to keep insulation from sliding according to manufacturer's written instructions.
- 3.4 INSTALLATION OF INSULATION FOR MISC. CONSTRUCTION
- A. Apply insulation units to substrates by method indicated, complying with manufacturer's written instructions. If no specific method is indicated then lay loose on surfaces.
 - B. Glass-Fiber Blanket Insulation: Install as specified on the drawings around Mechanical units, ceiling hatches, and along attic walkways.
 1. Maintain 3-inch clearance of insulation around recessed lighting fixtures not rated for or protected from contact with insulation.
 - C. Miscellaneous Voids: Install insulation in miscellaneous voids and cavity spaces where required to prevent gaps in insulation.

END OF SECTION 07210

SECTION 09512 - ACOUSTICAL TILE CEILINGS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Acoustical tiles and grid for drop-ceilings.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples: For each exposed product and for each color and texture specified, 6-inches- in size.
- C. Samples for Verification: For each component indicated and for each exposed finish required, prepared on Samples of size indicated below.
 - 1. Acoustical Tile: Two 6-inch by 6-inch samples.
 - 2. Exposed Moldings and Trim: Set of 6-inch- long Samples of each type and color.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Deliver acoustical tiles, suspension-system components, and accessories to Project site in original, unopened packages and store them in a fully enclosed, conditioned space where they will be protected against damage from moisture, humidity, temperature extremes, direct sunlight, surface contamination, and other causes.
- B. Before installing acoustical tiles, permit them to reach room temperature and a stabilized moisture content.

- C. Handle acoustical tiles carefully to avoid chipping edges or damaging units in any way.

1.5 FIELD CONDITIONS

- A. Environmental Limitations: Do not install acoustical tile ceilings until spaces are enclosed and weatherproof, wet work in spaces is complete and dry, work above ceilings is complete, and ambient temperature and humidity conditions are maintained at the levels indicated for Project when occupied for its intended use.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Seismic Performance: Acoustical ceiling shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
- B. Surface-Burning Characteristics: Comply with ASTM E 84; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
 - 1. Flame-Spread Index: Comply with ASTM E 1264 for Class B materials.
 - 2. Smoke-Developed Index: 450 or less.
- C. All material shall be asbestos free.

2.2 ACOUSTICAL TILES, GENERAL

- A. Low-Emitting Materials: Acoustical tile ceilings shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."
- B. Source Limitations:
 - 1. Acoustical Ceiling and Suspension System Tile: Obtain each type from single source from single manufacturer.

C. Acoustical Tile Standard: Provide manufacturer's standard tiles of configuration indicated that comply with ASTM E 1264 classifications as designated by types, patterns, acoustical ratings, and light reflectances unless otherwise indicated.

1. Mounting Method for Measuring NRC: Type E-400; plenum mounting in which face of test specimen is 15-3/4 inches away from test surface according to ASTM E 795.

2.3 ACOUSTICAL TILES

A. Manufacturers: Subject to compliance with requirements, provide products by the following:

1. USG Interiors, Inc.; Subsidiary of USG Corporation
2. Or approved equal.

B. Classification: Provide tiles complying with ASTM E 1264 Type III for type, form 4, and pattern E as follows:

1. Model: Form414 Frost ClimaPlus - for 15/16 inch grid profile DXL24HRC or approved equal.

C. Color: White.

D. LR: Not less than 0.75.

E. NRC: Not less than 0.65.

F. CAC: Not less than 36.

G. Edge/Joint Detail: SLB.

H. Thickness: 3/4 inch.

I. Modular Size: 24 inches by 24 inches.

J. Broad Spectrum Antimicrobial Fungicide and Bactericide Treatment: Provide acoustical tiles treated with manufacturer's standard antimicrobial formulation that inhibits fungus, mold, mildew, and gram-positive and gram-negative bacteria and showing no mold, mildew, or bacterial growth when tested according to ASTM D 3273 and evaluated according to ASTM D 3274 or ASTM G 21.

2.4 METAL SUSPENSION SYSTEMS, GENERAL

- A. Metal Suspension-System Standard: Provide manufacturer's standard metal suspension systems of types, structural classifications, and finishes indicated that comply with applicable requirements in ASTM C 635/C 635M.
- B. Attachment Devices: Size for five times the design load indicated in ASTM C 635/C 635M, Table 1, "Direct Hung," unless otherwise indicated. Comply with seismic design requirements.
- C. Wire Hangers, Braces, and Ties: Provide wires complying with the following requirements:
 - 1. Zinc-Coated, Carbon-Steel Wire: ASTM A 641/A 641M, Class 1 zinc coating, soft temper.
 - 2. Size: Select wire diameter so its stress at three times hanger design load (ASTM C 635/C 635M, Table 1, "Direct Hung") will be less than yield stress of wire, but provide not less than 0.106-inch- diameter wire.

2.5 METAL SUSPENSION SYSTEM

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. USG Interiors, Inc.; Subsidiary of USG Corporation.
 - 2. Or approved equal
- B. Direct-Hung, Double-Web Suspension System: Main and cross runners roll formed from and capped with cold-rolled steel sheet, prepainted, electrolytically zinc coated, or hot-dip galvanized according to ASTM A 653/A 653M, G30 coating designation.
 - 1. Structural Classification: Intermediate-duty system.
 - 2. Access: Upward, with initial access openings of size indicated below and located throughout ceiling within each module formed by main and cross runners, with additional access available by progressively removing remaining acoustical tiles.

2.6 METAL EDGE MOLDINGS AND TRIM

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. USG Interiors, Inc.; Subsidiary of USG Corporation.
 - 2. Or approved equal
- B. Roll-Formed, Sheet-Metal Edge Moldings and Trim: Type and profile indicated or, if not indicated, manufacturer's standard moldings for edges and penetrations complying with seismic design requirements; formed from sheet metal of same material, finish, and color as that used for exposed flanges of suspension-system runners.
 - 1. Provide manufacturer's standard edge moldings that fit acoustical tile edge details and suspension systems indicated and that match width and configuration of exposed runners unless otherwise indicated.
 - 2. For circular penetrations of ceiling, provide edge moldings fabricated to diameter required to fit penetration exactly.

2.7 MISCELLANEOUS MATERIALS

- A. Staples: 5/16-inch- long, divergent-point staples.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, including structural framing and substrates to which acoustical tile ceilings attach or abut, with Installer present, for compliance with requirements specified in this and other Sections that affect ceiling installation and anchorage and for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Examine acoustical tiles before installation. Reject acoustical tiles that are wet, moisture damaged, or mold damaged.

- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Measure each ceiling area and establish layout of acoustical tiles to balance border widths at opposite edges of each ceiling. Avoid using less-than-half-width tiles at borders, and comply with layout shown on reflected ceiling plans.

3.3 INSTALLATION OF SUSPENDED ACOUSTICAL TILE CEILINGS

- A. General: Install acoustical panel ceilings to comply with ASTM C 636/C 636M and seismic design requirements indicated, according to manufacturer's written instructions and CISCA's "Ceiling Systems Handbook."
- B. Suspend ceiling hangers from building's structural members and as follows:
 - 1. Install hangers plumb and free from contact with insulation or other objects within ceiling plenum that are not part of supporting structure or of ceiling suspension system.
 - 2. Splay hangers only where required to miss obstructions; offset resulting horizontal forces by bracing, countersplaying, or other equally effective means.
 - 3. Where width of ducts and other construction within ceiling plenum produces hanger spacings that interfere with location of hangers at spacings required to support standard suspension-system members, install supplemental suspension members and hangers in form of trapezes or equivalent devices.
 - 4. Secure wire hangers to ceiling suspension members and to supports above with a minimum of three tight turns. Connect hangers directly either to structures or to inserts, eye screws, or other devices that are secure and appropriate for substrate and that will not deteriorate or otherwise fail due to age, corrosion, or elevated temperatures.
 - 5. Do not attach hangers to deck. Attach hangers to structural members.
 - 6. Space hangers not more than 48 inches o.c. along each member supported directly from hangers unless

- otherwise indicated; provide hangers not more than 8 inches from ends of each member.
7. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by referenced standards and publications.
- C. Install edge moldings and trim of type indicated at perimeter of acoustical tile ceiling area and where necessary to conceal edges of acoustical tiles.
1. Apply acoustical sealant in a continuous ribbon concealed on back of vertical legs of moldings before they are installed.
 2. Screw attach moldings to substrate at intervals not more than 16 inches o.c. and not more than 3 inches from ends, leveling with ceiling suspension system to a tolerance of 1/8 inch in 12 feet. Miter corners accurately and connect securely.
 3. Do not use exposed fasteners, including pop rivets, on moldings and trim.
- D. Install suspension-system runners so they are square and securely interlocked with one another. Remove and replace dented, bent, or kinked members.
- E. Install acoustical tiles in coordination with suspension system and exposed moldings and trim. Place splines or suspension-system flanges into kerfed edges so tile-to-tile joints are closed by double lap of material.
1. Fit adjoining tile to form flush, tight joints. Scribe and cut tile for accurate fit at borders and around penetrations through tile.
 2. Hold tile field in compression by inserting leaf-type, spring-steel spacers between tile and moldings, spaced 12 inches o.c.
 3. Protect lighting fixtures and air ducts to comply with requirements indicated for fire-resistance-rated assembly.
- 3.4 CLEANING
- A. Clean exposed surfaces of acoustical tile ceilings, including trim and edge moldings. Comply with manufacturer's written instructions for cleaning and touchup of minor finish damage. Remove and replace tiles and other ceiling components that cannot be successfully

cleaned and repaired to permanently eliminate evidence of damage.

3.5 EXTRA MATERIALS

A. Provide 10 extra ceiling tiles to the field station for future replacement needs.

SCHEDULE OF ACOUSTICAL TILE CEILINGS AND HEIGHTS ABOVE FINISH FLOOR (AFF)

<u>ROOM #</u>	<u>CEILING HEIGHT</u>
ALL ROOMS	8.5' (AFF)

END OF SECTION 09512

PART 1 - GENERAL

1.1 SUMMARY:

A. This Section includes the following:

1. Interior lighting fixtures.
2. Lighting fixture supports.

1.2 SUBMITTALS

A. Product Data: For each type of lighting fixture, arranged in order of fixture designation. Include data on features, accessories, finishes.

1.3 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

B. Comply with NFPA 70.

PART 3 - PRODUCTS

2.1 MANUFACTURERS

A. The Interior Lighting is specified on the drawing but shall meet the following requirements for product selection:

1. Basis-of-Design Product: **Lithonia Lighting Contractor Select STACK 48"x24" and 24"x24" integrated LED Troffer with adjustable color temperature. Model STAKS 2X4 ALO6 SWW7 AND STAKS 2X2 ALO3 SWW7.**
2. Subject to compliance with requirements, provide either the named product or a comparable product by an approved equal.
3. **Include accessories for acoustical ceiling mounting, Self-Diagnostic LED emergency battery pack (1 light per room), and disconnect plug**

2.2 LIGHTING FIXTURES AND COMPONENTS, GENERAL REQUIREMENTS

- A. Recessed Fixtures: Comply with NEMA LE 4 for ceiling compatibility for recessed fixtures.
- B. Metal Parts: Free of burrs and sharp corners and edges.
- 4. Sheet Metal Components: Steel, unless otherwise indicated. Form and support to prevent warping and sagging.
- 5. Doors, Frames, and Other Internal Access: Smooth operating, free of light leakage under operating conditions, and designed to permit relamping without use of tools. Designed to prevent doors, frames, lenses, diffusers, and other components from falling accidentally during relamping and when secured in operating position.
 - f. Reflecting surfaces shall have minimum reflectance as follows, unless otherwise indicated:
 - a. White Surfaces: 85 percent.
 - b. Specular Surfaces: 83 percent.

2.4 LAMPS

- A. LED lamps can either be integrated LED or replaceable Lamps: Rated 45-55 W, 120V, nominal length 48 inches, 4000-6000 lumens, CRI 80, color temperature 4000 K, and average rated life 60,000 hours, unless otherwise indicated.

2.5 LIGHTING FIXTURE SUPPORT COMPONENTS

- A. Wires: ASTM A 641/A 641M, Class 3, soft temper, zinc-coated steel, 12 Gage.
- B. Rod Hangers: 3/16-inch minimum diameter, cadmium-plated, threaded steel rod.
- C. Hook Hangers: Integrated assembly matched to fixture and

line voltage and equipped with threaded attachment, cord, and locking-type plug.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Lighting fixtures: Set level, plumb, and square with ceilings and walls. Install lamps in each fixture.
- B. Comply with NFPA 70 for minimum fixture supports.
- C. Adjust aimable lighting fixtures to provide required light intensities.
- D. Connect wiring with flexible metal whip or according to local building codes.

(END OF SECTION 16500)

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